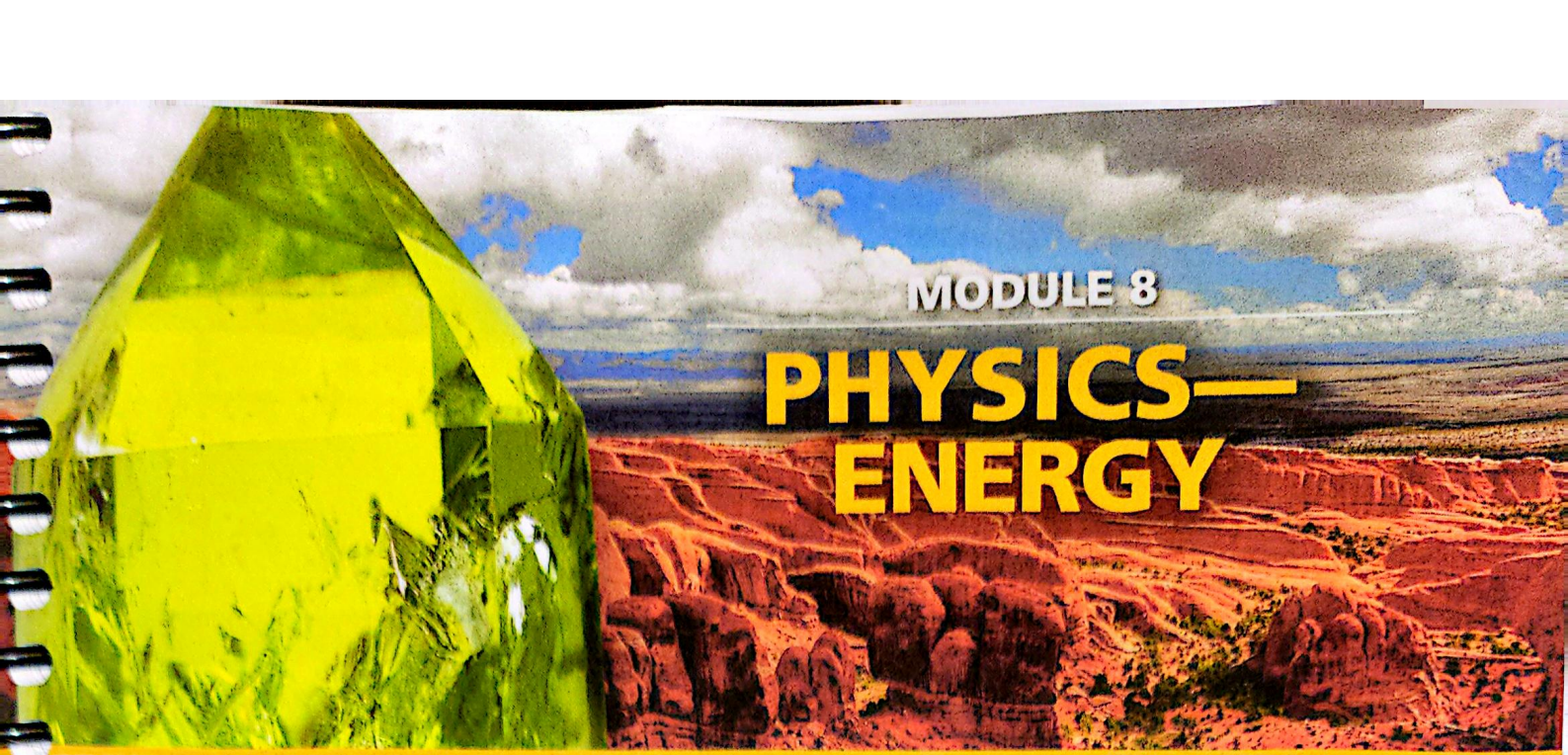




MODULE 8

PHYSICS— ENERGY

NOTETAKING

Take a few minutes to look back through your notebook so far. Have your notes been helpful? Are there any areas where you still need to concentrate? You've passed the halfway point through your Physical Science course, and now is the time to perfect your note taking skills. Some students find it helpful to use different colors while writing their notes to denote different concepts. Others use highlighters to make particular concepts stand apart. Consider experimenting with these ideas as you take notes in this module. Perhaps one of these methods will be helpful.



WHAT TO DO

Week 16, Day 3

- ☐ **Read** pages 271–275, the introduction and Energy section, including the subsections: Types of Energy: Kinetic Energy.
- ☐ **Take notes** using the following prompts and write any thoughts or questions you have on the Personal Notes page.
- ☐ **Write** the definitions to any vocabulary words you come to in the space provided.
- ☐ **Answer On Your Own** questions 8.1–8.2 and check your answers.
- ☐ **Check off** this day on your Daily Schedule in the front of this notebook.

ENERGY

VOCABULARY

Energy—

Note Prompt	Notes:
What does the text say energy is defined as in addition to the definition you wrote above?	In science, energy is defined as the _____ _____ in some way.
What is the SI unit for energy? What is it equal to?	



As described in the text, explain where the energy came from and how it is transferred as the woman hits the ball with the racket.



Types of Energy: Kinetic Energy

What does the text say the kinetic energy of any moving object depends on?

What is the equation for finding kinetic energy (KE)?



Complete the problem-solving steps for Example 8.1 as you read through it in your textbook.

A.	Step 1: List the knowns and unknown.	
	Step 2: Write out the equation you will use.	
	Steps 3 and 4: Write out the equation, substituting the numbers from the list in step 1. Solve the problem.	

B.	Step 1: List the knowns and unknown.	
	Step 2: Write out the equation you will use.	
	Steps 3 and 4: Write out the equation, substituting the numbers from the list in step 1. Solve the problem.	
C.	Step 1: List the knowns and unknown.	
	Step 2: Write out the equation you will use.	

C.

Steps 3 and 4:

Write out the equation, substituting the numbers from the list in step 1. Solve the problem.

8.1 ON YOUR OWN

How is the kinetic energy of an object determined?

**8.2 ON YOUR OWN**

A 75.0 kg man is jogging at a speed of 3.5 m/s. What is his kinetic energy?





WHAT TO DO

Week 16, Day 4

- ☐ **Read** pages 275–279, subsection: Types of Energy: Potential Energy.
- ☐ **Take notes** using the following prompts and write any thoughts or questions you have on the Personal Notes page.
- ☐ **Write** the definitions to any vocabulary words you come to in the spaces provided.
- ☐ **Answer** On Your Own questions 8.3–8.5 and check your answers.
- ☐ **Check off** this day on your Daily Schedule in the front of this notebook.

Types of Energy: Potential Energy

VOCABULARY

Potential Energy—

Note Prompt	Notes:
What examples of potential energy does the text give you? Can you think of any others to add to your notes?	
What are the two forms of potential energy?	
What is the equation for determining gravitational potential energy? (Equation 8.2 in your textbook)	



Complete the problem-solving steps for Example 8.2A as you read through it in your textbook.

Step 1: List the knowns and unknown.	
Step 2: Write out the equation you will use.	
Steps 3 and 4: Write out the equation, substituting the numbers from the list in step 1. Solve the problem.	

Give a couple examples of objects that have gravitational potential energy.

Give a couple examples of objects that have elastic potential energy.

8.3 ON YOUR OWN

what factors determine the gravitational potential energy of an object?

**8.4 ON YOUR OWN**

what type of energy does a compressed pogo stick (Figure 8.9) have?

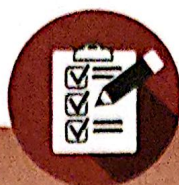


FIGURE 8.9
Pogo Stick

**8.5 ON YOUR OWN**

A 55.00 kg stunt man walks up the steps from the ground to the roof of a 72.8 m tall building. How much gravitational potential energy does he have at the top of the building?





WHAT TO DO

Week 17, Day 1

- ☐ **Complete** Experiment 8.1 on pages 279–280 of the text (turn to the lab report form in the lab section of this notebook).
- ☐ **Read** pages 280–282, the Forms of Energy: Mechanical Energy subsection.
- ☐ **Take notes** using the following prompts and write any thoughts or questions you have on the Personal Notes page.
- ☐ **Explore** the You do Science activity.
- ☐ **Check off** this day on your Daily Schedule in the front of this notebook.

Forms of Energy: Mechanical Energy

What does the text say **mechanical energy** is?



YOU DO SCIENCE: BALL BOUNCE

DATA TABLE I		
	Height of bounce (m)	Observations
Basketball		
Tennis ball		

DATA TABLE I, continued

	Height of bounce (m)	Observations
Basketball (with tennis ball)		
Tennis ball (with basketball)		

How can you explain what happened in terms of mechanical energy—potential and kinetic energy?



WHAT TO DO

Week 17, Day 2

- ☐ **Read** pages 282–287, the Types of Energy: Thermal Energy, Chemical Energy, Electrical Energy, Electromagnetic Energy, Sound Energy, Nuclear Energy, and Conservation of Energy subsections.
- ☐ **Write** the definitions to any vocabulary words you come to in the spaces provided.
- ☐ **Take notes** using the following prompts and write any thoughts or questions you have on the Personal Notes page.
- ☐ **Answer On Your Own** questions 8.6–8.7 and check your answers.
- ☐ **Check off** this day on your Daily Schedule in the front of this notebook.

Types of Energy, continued



As you read each subsection, write down the notes you think are important to remember. What is different about each type of energy? What is similar about each type? Don't forget to include anything in blue.

Thermal energy

Chemical energy

Electrical energy

Electromagnetic energy

Sound energy

Nuclear energy

Conservation of Energy

What is meant by **energy conversion**?

VOCABULARY

Law of conservation of energy—

What example of conservation of energy is given in text? What is meant by a **closed system**?

Explain some of the everyday examples of energy conversions you read about on pages 286–287.

8.6 ON YOUR OWN

Identify the form of energy in each of the following examples.

Example	Form of Energy
a. heating a pot of water over a fire	
b. gasoline as fuel for a car	
c. lightning	
d. a bouncing ball	
e. the vibrations of a beating drum	
f. sunlight	

**8.7 ON YOUR OWN**

what does the law of conservation of energy state?





WHAT TO DO

Week 17, Day 3

- ☐ **Read** pages 288–293, the Energy and Work section, including the Calculating Work, Power, and Calculating Power subsections.
- ☐ **Take notes** using the following prompts and write any thoughts or questions you have on the Personal Notes page.
- ☐ **Write** the definitions to any vocabulary words you come to in the spaces provided.
- ☐ **Answer** On Your Own questions 8.8–8.10 and check your answers.
- ☐ **Check off** this day on your Daily Schedule in the front of this notebook.

ENERGY AND WORK

VOCABULARY

Work—

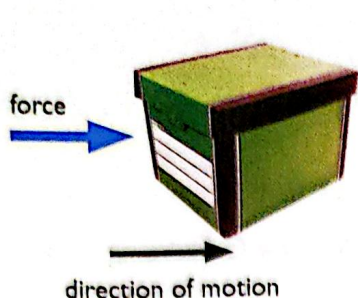
What are the two conditions that must be met in order for work to be done (according to science)?

1.

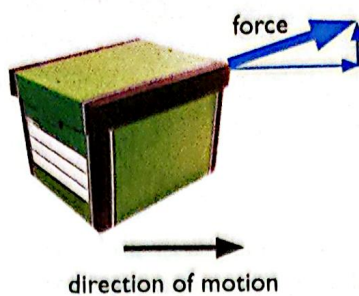
2.



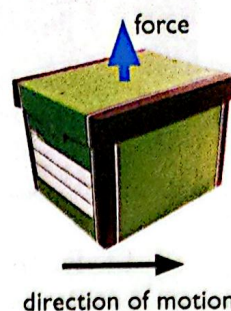
Circle the forces (or parts of a force) that are doing work.



A Pushing force and motion in same direction



B Part of pulling force in same direction of motion



C Lifting force not in direction of motion

Calculating Work

What is the equation for work?

What are the units for work?
What else uses these units?



Complete the problem-solving steps for Example 8.3 as you read through it in your textbook.

Step 1: List the knowns and unknown.	
Step 2: Write out the equation you will use.	

Steps 3 and 4:

Write out the equation, substituting the numbers from the list in step 1. Solve the problem.

Power**VOCABULARY**

Power—

What are the two ways you can increase power?

Calculating Power

What is the equation for power?

What are the units for power?
What is it equal to?



Complete the problem-solving steps for Example 8.4 as you read through it in your textbook.

A.	Step 1: List the knowns and unknown.	
	Step 2: Write out the equation you will use.	
	Steps 3 and 4: Write out the equation, substituting the numbers from the list in step 1. Solve the problem.	
B.	Step 1: List the knowns and unknown. Check units and convert if needed.	

B.	Step 1B: Convert.	
	Step 2: Write out the equation you will use.	
	Steps 3 and 4: Write out the equation, substituting the numbers from the list in step 1. Solve the problem.	

8.8**ON YOUR OWN**

What conditions are required for a force to do work on an object?



8.9 ON YOUR OWN

A book is sitting on a table. The table is exerting an upward force on the book to support it as it rests on the table. Does this force do work? Explain.

**8.10 ON YOUR OWN**

How much work (in joules) do you do when you use a 25 N force to lift a bag of potting soil from the floor to a shelf 1.5 m high? If you lift the bag in 2.0 s, how much power (in watts) do you generate?





WHAT TO DO

Week 17, Day 4

- ☐ **Complete** Experiment 8.2 on pages 294–295 of the text (turn to the lab report form in the lab section of this notebook).
- ☐ **Check off** this day on your Daily Schedule in the front of this notebook.



WHAT TO DO

Week 18, Day 1

- ☐ **Read** pages 296–304, the Work and Machines section, including the Work Input and Work Output, the Mechanical Advantage, and the Simple Machines subsections.
- ☐ **Take** notes using the following prompts and write any thoughts or questions you have on the Personal Notes page.
- ☐ **Write** the definitions to any vocabulary words you come to in the spaces provided.
- ☐ **Answer** On Your Own questions 8.11–8.12 and check your answer.
- ☐ **Check off** this day on your Daily Schedule in the front of this notebook.

WORK AND MACHINES

**What three things can a machine do (change) to make work easier?
What example does the text give for each?**

1.

Example:

2.

Example:

3.

Example:

Work Input and Work Output

VOCABULARY

Input force—

Output force—

Input distance—

Output distance—

Mechanical Advantage

What is a
machine's **actual
mechanical
advantage** (AMA)?



What is the equation for AMA?	
What is a machine's ideal mechanical advantage (IMA) ?	What is the equation to find IMA?
↓	↓



Complete the problem-solving steps for Example 8.5 as you read through it in your textbook.

A.	Step 1: List the knowns and unknown.	
	Step 2: Write out the equation you will use.	

A.

Steps 3 and 4: Write out the equation, substituting the numbers from the list in step 1. Solve the problem.

B.

Step 1: List the knowns and unknown. Check units and convert if needed.

Step 2: Write out the equation you will use.

Steps 3 and 4: Write out the equation, substituting the numbers from the list in step 1. Solve the problem.

Most machines have a mechanical advantage greater than 1 ($IMA > 1$).
What does each of the following mechanical advantages mean?

$IMA > 1$

$IMA = 1$

$IMA < 1$

What is efficiency when referring
to a machine?

What is the equation for efficiency?

Simple Machines



For each of the simple machines you read about in Infographic 8.1 in your text, briefly describe it, give an example, and list whether the IMA is greater than, less than, or equal to 1.

Simple Machine	Description	Example	IMA
Inclined plane			

Simple Machine	Description	Example	IMA
Wedge			
Pulley			
Wheel and axle			
Lever	1. 2. 3.		
Screw			

8.11 ON YOUR OWN

why is the actual mechanical advantage of a machine always less than its ideal mechanical advantage? And why can no machine be 100% efficient?

**8.12 ON YOUR OWN**

what is the ideal mechanical advantage of a ramp if its length is 4.0 m and its higher end is 0.5 m above its lower end?





WHAT TO DO

Week 18, Day 2

Now that you've read the module, taken notes, and completed all the On Your Own questions, it's time to study! To do that, begin by completing the checklist below.

- ☐ **Choose** one of the experiments you conducted in this module and using the full lab report form in the lab section of this notebook, **write** your full lab report.
- ☐ Before you start the study process, you might want to take a moment and think about all that you've learned in this module. Do you view the world differently than you did before reading it? *If you like*, **write down** your thoughts and questions on the Personal Notes page.
- ☐ **Prepare for the exam** by reviewing your notebook pages. What does this mean? Well, take some time to look at all your notes and review what you have learned so far.
- ☐ **Answer** the Study Guide questions—it works best if you first try to do this without “looking back” at your notes and textbook. When you are done, you can use your text and notes to fill in any answers you did not know.
- ☐ **Check** your answers (or have your parent check your answers) with the *Solutions and Tests Manual*. **Review** and fix anything you got wrong and reread parts of the text if needed.
- ☐ **Check off** this day on your Daily Schedule in the front of this notebook.

STUDY GUIDE FOR MODULE 8

1 Match the term to the correct definition.

- | | |
|---------------------|---|
| a. Energy | the force exerted by a machine |
| b. Potential energy | the distance the output force is exerted through, or how far the load moves |
| c. Kinetic energy | the distance the input force is applied |
| d. Input force | the ability to do work |
| e. Output force | energy that is stored as a result of position or shape |
| f. Input distance | the force a person exerts on a machine |
| g. Output distance | the energy of an object due to its motion |

2 If the speed of an object doubles, what happens to its kinetic energy?

- | | |
|----------------------|------------------|
| a. KE doubles | c. KE is halved |
| b. KE stays the same | d. KE quadruples |

3 Which of the following is an example of an object with elastic potential energy?

- | | |
|-----------------------|---------------------|
| a. a stretched spring | c. a moving arrow |
| b. books on a shelf | d. a falling oyster |

4 Which of the following is an example of electromagnetic energy?

- | | |
|-------------------|--------------------------|
| a. a speeding car | c. a falling stone |
| b. sunlight | d. a stretched bowstring |

5 Friction causes kinetic energy to be converted into what?

- | | |
|----------------------|--------------------|
| a. potential energy | c. chemical energy |
| b. mechanical energy | d. thermal energy |

6 What is the energy stored in the bonds between atoms and molecules called?

- a. kinetic energy
- b. mechanical energy
- c. chemical energy
- d. thermal energy

7 Which of the following is an example of the conversion of gravitational potential energy to kinetic energy?

- a. striking a match
- b. kicking a soccer ball
- c. a falling raindrop
- d. a gasoline-powered engine

8 What does the law of conservation of energy state?

9 What is work?

- a. the product of speed and force
- b. the sum of power and force
- c. the product of force and distance
- d. the sum of motion and force

10 Which of the following is a unit of work?

- a. joule
- b. watt
- c. horsepower
- d. newton

11 Power is equal to work divided by which of the following variables?

- a. time
- b. force
- c. distance
- d. mechanical advantage

12 How is the work output of a machine related to its work input?

- a. work output is always less than work input
- b. work output is always greater than work input
- c. work output is about the same as work input
- d. work output is always zero

13 What can you say about the output force of a machine with a mechanical advantage much larger than 1?

- a. the output force is much less than its input force
- b. the output force is much larger than its input force
- c. the output force and input force are about the same
- d. the output force is in the same direction as its input force

14 What other type of simple machine can a screw be considered a type of?

- a. lever
- b. inclined plane
- c. pulley
- d. wedge

15 Does a weightlifter do work on the weight bar as he lifts it over his head? Is work done on the weight bar as he stands still holding the weights over his head for 5 seconds? Explain.

- 16** A small meteoroid is approaching Earth. It has a mass of 120.0 kg and a speed of 20.0 m/s. How much kinetic energy does the meteoroid have?

- 17** Suppose 200.0 kg of building supplies are lifted in the air to be placed onto a workstation. How much gravitational potential energy is needed to lift the supplies 3.00 meters into the air?

18

A machine has an efficiency of 75%. What happens to 75% of the work put into the machine, and what happens to the other 25%?

19

Say you are cracking nuts with a nutcracker.

- a. You apply an input force of 12.5 N to the nutcracker while the output force is 50.0 N. What is the actual mechanical advantage of the nutcracker?

- b. If the input distance of the nutcracker is 15.0 cm and the output distance is 3.0 cm. What is the ideal mechanical advantage of the nutcracker?

20

You exert a force of 500.0 N to walk 4.0 m up a flight of stairs in 4.0 s. How much power do you use?



WHAT TO DO

Week 18, Day 3

- ☐ Take about 20 minutes to **review for the exam** (review your notes in this notebook and the Study Guide questions).
- ☐ **Take the exam** (it can be found in the *Solutions and Test Manual*). You should take the exam without your notes or your textbook.
- ☐ With a parent, **check your test answers** against the answers in the *Solutions and Tests Manual*.
- ☐ Go back in the text and **review anything you got wrong** on the test.
- ☐ **Check off** this day on your Daily Schedule in the front of this notebook.